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Leveraging Retrieval-Augmented Generation to Support Legal Question Answering: Evidence from Indonesian Human Trafficking Court Decisions

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***Abstract*—The exponential growth of court decisions in Indonesia has posed a crucial challenge for legal practitioners in obtaining relevant information. Conventional search systems fail to capture the in-depth legal context, while Large Language Models (LLMs) are prone to producing hallucinations that can mislead legal reasoning. This study proposes and tests the implementation of Retrieval-Augmented Generation (RAG) to support Legal Question Answering LLM-based Quality Assurance (LQA) to improve factual accuracy. This study used 408 Indonesian court decisions related to criminal cases. Human trafficking data collected and standardized from 143 district courts. The RAG framework is designed in three stages: indexing, search, and incremental generation. We evaluated three Open-source LLM models: Gemma, LLaMA, and Qwen. Three models are also combined with two retrieval methods: BM25 (lexical) and Dense (semantics). Experimental results show that Qwen 3, especially when combined with BM25 RAG, consistently produces the highest overall answer quality across all evaluation metrics (ROUGE and BLEU). The BM25 method is significantly more effective than dense retrieval. Due to the highly standardized nature of court decision documents, Qwen demonstrated peak performance on structured information categories such as "*identitas_terdakwa*", achieving a ROUGE-L score of 0.899. The primary contribution of this study is demonstrating that RAG significantly reduces the risk of LLM hallucinations in a legal context. However, error analysis suggests that future improvements should focus on strengthening generation controls to address these issues: unsupported generation and remaining model-rejection behavior.**

***Keywords*—Retrieval augmented generation; human trafficking; law; large language models; information retrieval; question answering.**

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I. INTRODUCTION

The increasing number of court decision documents in Indonesia poses new challenges for practitioners, academics, and researchers in the legal field in identifying relevant precedents to support legal reasoning. Although traditional Information Retrieval (IR) can search documents based on keywords, IR often fails to capture the semantic and contextual elements of legal texts [1], [2], [3]. The entire decision-analysis process at the Judicial Commission of the Republic of Indonesia (KYRI) is currently conducted manually. This process involves universities and non-governmental organizations. As a result, an average of only 100 decisions can be analyzed per year. This is significantly fewer than the number of court decisions listed in the Supreme Court's decision directory (https://putusan3.mahkamahagung.go.id/). There were 860,683 decisions recorded in 2023 and 822,242 in 2024. This equates to approximately 800,000 decisions per year. The large number of available court decisions is not only a challenge but also a valuable source of knowledge for legal professionals. Legal Question Answering (LQA) is one form of utilizing the vast knowledge sourced from court decisions. LQA can help identify precedents in new cases [4] or in the summarization of legal texts [5].

Nevertheless, the reliance on high-quality training data and the substantial cost of annotation remain significant obstacles to broader adoption. This challenge highlights the pressing

need for LQA datasets and models that can expedite the development of production-ready applications in critical domains, such as law [6].

Referring to research in the field of Large Language Models (LLM), LLM has shown remarkable achievements [7]. LLM allows users to ask questions in a legal context in natural language and obtain accurate answers [8]. The capabilities of LLMs enable models to extract important information from legal texts and present it in a more accessible form to support decision-making. As a result, transformer-based models such as GPT [9], [10] and BERT [11] are quite popular in the field of research on the use of AI in the legal field. Antoine et al. introduced the Long-form Legal Question Answering (LLeQA) dataset, which was designed to train LLMs to answer legal questions [6], although they did not develop LLMs specifically for answering legal questions.

Despite the impressive performance of the LLM, it remains a seemingly inaccessible black box, which can lead to hallucinations. In the legal realm, hallucinations mean that the model can produce seemingly confident but false, fabricated responses not based on legal facts. This problem becomes crucial when applying the LLM to the legal field because it can mislead legal reasoning [12]. The solution to this problem is the Retrieval-Augmented Generation (RAG) technique. RAG provides a solution by strengthening the output of LLM based on factual evidence [13], [14], [15]. RAG will improve document representation by embedding context that is continuously updated during model training. This mechanism will ensure better alignment between the generated output and the retrieved document. Chouhan et al. introduced the LexDrafter framework, leveraging RAGs and predefined term definitions to aid in drafting definition sections in legislative documents [16]. This framework harmonizes legal terminology across regulations, reduces human error, and improves the efficiency of the document drafting process [16]. This research focuses on the standardization of legal terminology, not on QA in the legal field.

This research aims to address the problem of LLM hallucinations in LQA. This problem has been a long-standing challenge in the field of artificial intelligence research in law. This research focuses on the functionality of the LLM for conducting LQA-based decision analysis. Based on previous research, there has been no research on using an LLM to answer questions about analyzing Indonesian legal decisions with supporting references. RAG was chosen because it does not require retraining the LLM with newly available data, thus reducing the computational burden and cost [17]. In addition to efficiency reasons, RAG also allows the LLM model to incorporate references, in this case sourced from court decision datasets, in constructing response answers. This feature increases the transparency of the AI model and allows users to cross-check by retrieving content for accuracy and contextual relevance. This study uses the Human Trafficking Crime Court Decision as the dataset [18]. The consideration for selecting this dataset is due to the complexity and the large number of human trafficking cases that have occurred recently.

By focusing on the legal domain, this study not only evaluates the effectiveness of RAG in reducing hallucinations but also demonstrates its potential to provide context-specific legal insights. This study also emphasizes a more comprehensive evaluation of the model's performance in answering legal questions. This study goes beyond conventional lexical overlap metrics and also considers the quality of the generated answers and the relevance of the supporting evidence obtained.

Answer quality is assessed using various indicators, including ROUGE, BLEU, token-level agreement, semantic similarity, linkage matching, and consistency checks for legal entities such as numbers, dates, and names. Furthermore, the ratio of unsupported sentences and repetitions is used as another indicator to check whether the model's answers remain grounded in the legal context and avoid redundant or unsupported output. This series of evaluations is intended to provide a more reliable assessment of factual accuracy, semantic relevance, and retrieval effectiveness in legal Q&A.

II. Materials and Method

Natural Language Processing (NLP) is a scientific discipline that studies communication between computers and humans using natural language—various related research. NLP has enabled the development of sophisticated natural language models, such as LLM. Model LLM has been trained by using a large text corpus, with the results that are impressive in tasks such as translation, summarization, and answering questions [19]. This makes LLM very effective for answering user questions. Integration RAG plays an important role in addressing hallucinations that often arise in LLM-generated answers. RAG is designed to increase LLM capacity by incorporating external knowledge sources as references for more accurate and verifiable answers. RAG bridges the gap between information and reasoning by combining information retrieval-based methods with a generative model.

As illustrated in Figure 1, the RAG framework in this study consists of three core stages: Indexing, Retrieval, and Generation. The data indexing stage will extract and convert the data to plain text, then segment it, encode it into vector representations, and store it in a vector database for efficient vector similarity-based search.

At this retrieval phase, a query from the user's data is also converted into a vector, and a similarity score with the indexed chunks is calculated. The top-K most relevant chunks are selected for the next stage. In this phase, generation, LLM produces a response by combining the query with the retrieved documents, either by leveraging prior knowledge or restricting the output to provided sources, depending on the specific requirements of the task [20]. This study also employs an empirical research methodology in the field of law, which includes announcements and observations at the Indonesian Judicial Commission. Purposive sampling was used to determine the interview and observation samples [21].

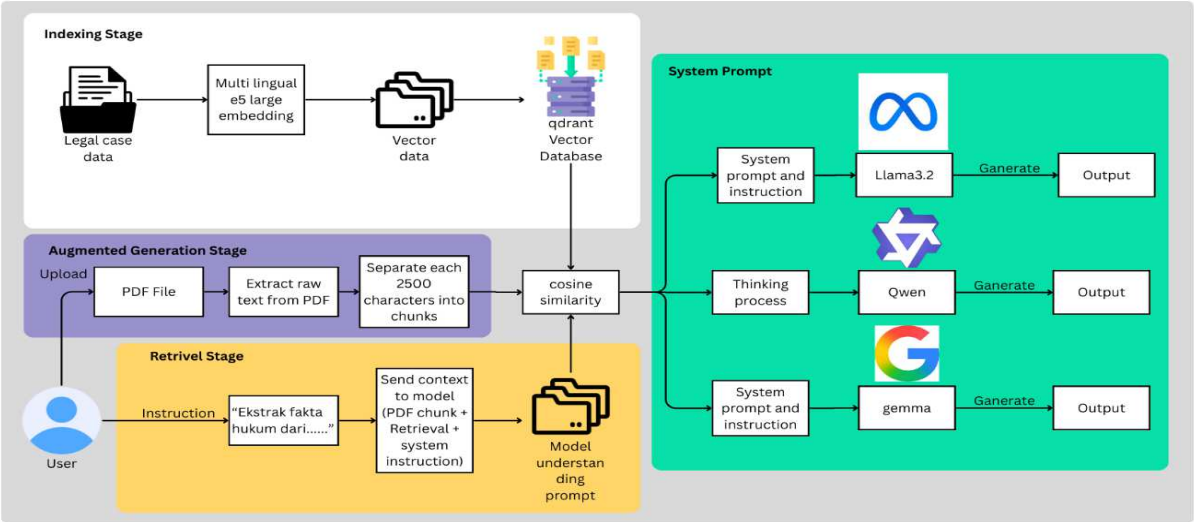


Fig. 1 RAG framework

A. *Dataset Construction*

Decision court document collected using web scraping techniques in Python, with sources from the official website of the Supreme Court Decision Directory of the Republic of Indonesia [18]. One type of case that requires special attention is human trafficking. As a complex transnational crime, human trafficking cases have broad legal and social dimensions. Therefore, the availability of comprehensive, structured court decision data that is relevant to computational needs is essential to support this research. According to previous research, human trafficking poses a serious threat to global health and security, although the availability of datasets on criminal cases involving human trafficking remains limited [22]. The data scraping process conducted on the Supreme Court's website is described in detail in Figure 2.

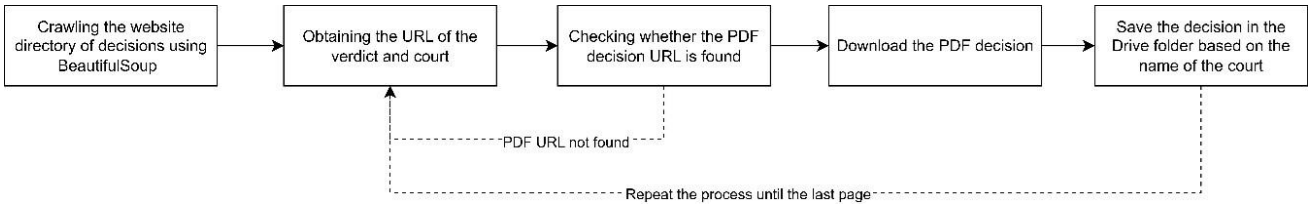


Fig. 2 Web crawling flow on the website of the Supreme Court of the Republic of Indonesia

Each case collected from the website consists of a complete court decision document in PDF format. Every document is stored in a Google Drive folder and labeled with the year and the name of the district court where the case was decided. The distribution of judgment sources across various district courts in Indonesia is one of the factors considered in case selection. A total of 143 district courts served as sources of judgments. Figure 3 shows the 10 district courts from which the most decisions were drawn during the 2023 to 2024 period.

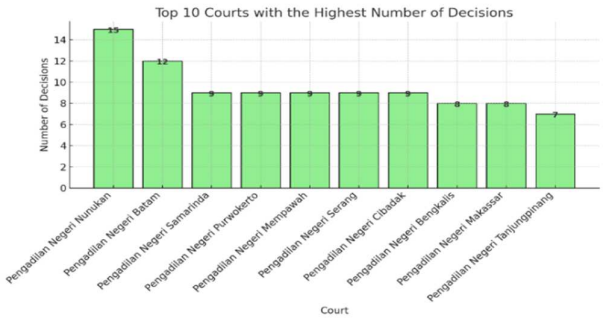


Fig. 3 Court document distribution from the largest to the smallest.

As an effort to ensure that the legal material is relevant and complete, each decision document will be filtered manually by Legal Experts. The selection process aims to evaluate whether a judgment contains the components hereinafter referred to as "features" required under Chief Justice of the Supreme Court Decision No. 359/KMA/SK/XII/2022 regarding templates and standards for the drafting of court judgments. Legal experts do this Evaluation to maintain validity and credibility. Dataset built based on the systematics of decisions [23]. Manual verification also aims to improve the reliability of LLM-based feature extraction in entity recognition and decision summarization. Through this expert validation approach, it is possible to combine rule-based, abductive, and case-oriented reasoning to ensure that the results of decision analysis are more adaptive and accurate.

This study has succeeded in obtaining 408 court decisions on criminal cases of human trafficking. 260 cases were adjudicated in 2023 and another 148 cases in 2024. The feature extraction workflow shown in Figure 4 indicates that, following initial filtering, 404 documents were criminal judgments related to human trafficking, while 4 were

classified as civil cases. All valid judgments were then extracted into 24 features, including: decision number, court name, district court name, case details, defendant information, and other relevant attributes.

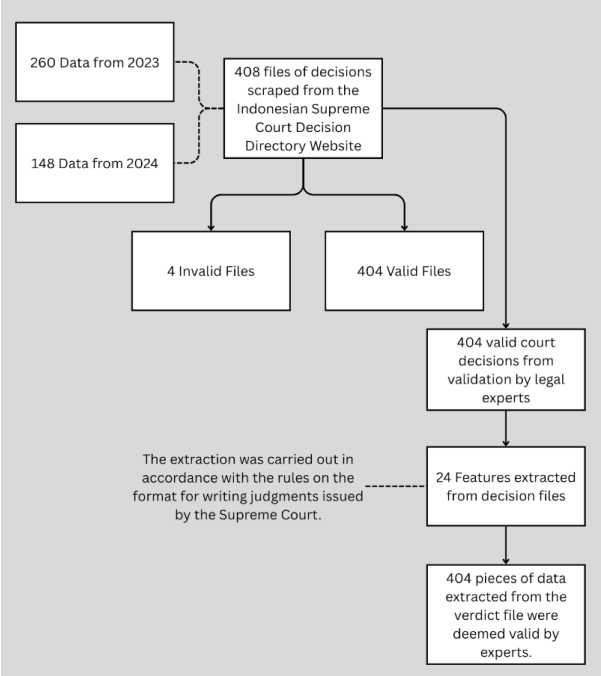


Fig. 4 Distribution of court decisions in special criminal cases involving human trafficking

B. Preprocessing

This study divides the preprocessing into three phases, as illustrated by the RAG Framework in Figure 1. The first phase, the indexing phase, transforms 404 court decision documents into vector representations for storage in a vector database. The second phase is preprocessing, where the decision documents are segmented into smaller parts to fit the LLM's capacity. The third phase is retrieval. In this phase, user queries are also converted into vectors so that the system can process them.

1) Indexing Stage:

At this stage, the court decision documents that experts have validated will be split into three parts: 80% for training, 10% for testing, and 10% for validation. Next, the court decision documents will be vectorized using the Transformer model (multilingual-e5-large-instruct), with a vector size of 1024 dimensions. Representation using numerical vectors via an embedding model aims to capture the semantic meaning of the text. After the 24 features from the judgment data have been successfully vectorized, the vector data will be stored in the vector database, Qdrant.

2) Augmented Generation Stage

At this stage, the content of the court decision document is chunked into several sections, each containing 2,500 characters. This chunking process aims to assess the similarity of metadata across sections of the court ruling document. To ensure the accuracy of the extraction results, metadata similarity is measured using cosine similarity. The selected sections are those with cosine similarity scores> 0.8.

3) Retrieval Stage:

At this stage, the instructions for extracting judgment fragments will be converted into vectors using multilingual embedding. These vectors will be used to calculate relevance to the metadata stored in the database vectors and will then be matched with other fragments that have been or will be extracted. This retrieval process is similar to how search engines work. The difference here is that the model does not rely solely on keyword matching, but also uses an LLM to interpret the user's query in natural language. The LLM also identifies the most relevant nodes in the vector database. The model ensures that the generated response is based on semantic understanding, not just simple keyword matching. [24].

C. Model Architecture

1) Gemma:

Gemma is an open-source LLM model released by Google DeepMind, designed with efficiency and flexibility in mind. Gemma is available in various sizes (ranging from 270 million to 27 billion parameters) and supports multimodal inputs (text and images) as well as long-context processing of up to 128,000 tokens. Gemma also supports over 35 languages natively and offers training in 140 languages. Given these features, Gemma is particularly well-suited for handling lengthy and complex court rulings, which often contain multilingual elements.

Across various metrics, Gemma 3 demonstrates competitive performance in reasoning, following instructions, and multilingual tasks. Another strength lies in its ability to call functions and generate structured output. Gemma also facilitates integration with the RAG framework to develop LQA systems. However, like other LLMs, Gemma 3 remains prone to hallucinations, requiring retrieval-based augmentation to ensure the factual accuracy and reliability of its outputs [25].

2) LLaMA:

The LLaMA model is available in various parameter sizes, ranging from lightweight variants optimized for resource-constrained devices to larger models capable of handling more complex processing tasks. This range allows researchers to select the most appropriate version based on their infrastructure capacity and specific use cases. Like Gemma, LlaMA 3.2 also supports long-context processing, making it highly relevant for legal texts, which often involve lengthy narratives and complex argumentative structures [26].

3) Qwen:

Qwen is an LLM developed by Alibaba Group. This open-source model is built for high performance and is intended for both research and practical applications. The latest generations, such as Qwen 3, have parameter counts ranging from 1.8 billion to tens of billions. These versions are designed to support a wide range of tasks, including reasoning, code generation, question answering, and long-form document processing. The Qwen 3-4B variant offers a balance between accuracy and computational efficiency, making it suitable for use in environments with moderate computational resources. This model also supports multilingual processing with long-context understanding

capabilities, making it particularly well-suited for analyzing long and complex legal documents [27].

In benchmark evaluations, Qwen demonstrates competitive performance compared to models of similar scale, particularly in natural language understanding, reasoning, and knowledge-intensive question answering. Another advantage of Qwen is its support for structured outputs and compatibility with RAG frameworks, both of which are crucial for ensuring factual accuracy in sensitive domains such as law. Nonetheless, like other LLMs, Qwen remains prone to hallucination, highlighting the need for integration with robust retrieval and verification mechanisms [28].

D. Evaluation

This study evaluates the model's performance using two approaches: 1) the quality of the generated answers [29] and 2) the quality of the supporting information obtained [30]. The quality of the answers is evaluated using measures of surface-level, semantic-level, and lexical overlap. In addition to ROUGE and BLEU, the evaluation also reports various indicators, including: Token F1, semantic similarity, relevance match, numerical consistency, date consistency, name consistency, unsupported sentence ratio, and repetition ratio. The F1 score measures the word-level similarity between the generated and reference responses'

Semantic similarity, on the other hand, measures whether the response conveys the same meaning. The consistency metric checks whether the accuracy of key legal information, such as numbers, dates, and names, is maintained in the generated response. The unsupported sentence ratio will be used to estimate whether the extracted context insufficiently supports the generated statement. Meanwhile, the repetition ratio identifies repetitive or degenerative output patterns.

1) Baseline System:

Four evaluation settings were compared: 1) Dense RAG using vector retrieval followed by generation; 2) BM25 RAG using lexical retrieval followed by generation; 3) Non-RAG LLM generating answers without retrieved context; and 4) BM25 Retrieval-Only returning retrieved evidence without language model generation [31], [32]. This baseline distinguishes between the contributions of information retrieval and information generation, and provides a direct comparison between semantic and lexical information retrieval.

2) Retrieval Metrics:

Retrieval performance is measured by Recall@1, Recall@3, Recall@5, Recall@10, Mean Reciprocal Rank (MRR), and normalized discounted cumulative gain at rank 10 (nDCG@10). Recall@k measures whether a relevant chunk appears within the first k retrieved chunks, MRR rewards earlier placement of the first relevant chunk, and nDCG@10 evaluates ranking quality across the top ten results [31], [32].

3) ROUGE:

The Recall-Oriented Understudy for Gisting Evaluation (ROUGE) metric is one of the most widely used methods for evaluating the quality of automatically generated text, particularly in tasks such as text summarization, machine translation, and other natural language generation (NLG) systems. ROUGE focuses on measuring the overlap between n-grams (i.e., sequences of words), word pairs, or sub-sequences in the generated output and those in the reference text. There are several variations of ROUGE, such as ROUGE-N (measuring n-gram overlap), ROUGE-L (based on the longest common subsequence), and ROUGE-S (co-occurrence of skip-bigrams) [33]. These variations allow researchers to capture different aspects of text similarity, including exact matches, sequence fluency, and semantic closeness. Using equation 1, ROUGE measures the n-gram overlap between the candidate text (system output) and the reference text (human writing), where, $gram_n$ is n-gram in the reference summary, and $Count_{match}$ is the number of overlapping n-grams between the candidate and reference.

$$ROUGE - N = \frac{\sum_{S \in RefSumm} \sum_{gram_n \in S} Count_{match}(gram_n)}{\sum_{S \in RefSumm} \sum_{gram_n \in S} Count(gram_n)} \quad (1)$$

4) BLEU:

BLEU operates by measuring the degree of n-gram overlap between a candidate translation and one or more human reference translations. BLEU uses a modified precision approach, where repeated words in the candidate are not counted beyond their frequency in the reference [34]. This helps prevent artificially inflated scores through word repetition. BLEU also incorporates a brevity penalty (BP) to prevent overly short translations that may achieve high precision but fail to convey the full meaning [35]. The general BLEU score is defined in Equation 2, while the modified n-gram score is presented in Equation 3.

$$BLEU = BP.\exp(\sum_{n=1}^{N} w_n.\log p_n) \quad (2)$$

$$p_n = \frac{\sum_{C \in Candidates} \sum_{gram_n \in C} Count_{clip}(gram_n)}{\sum_{C \in Candidates} \sum_{gram_n \in S} Count(gram_n)} \quad (3)$$

$gram_n$ in p_n is n-gram in the candidate translation and $Count_{clip}$ is number of times an n-gram appears in both candidate and reference, clipped to the maximum frequency in the reference (to avoid overcounting). The weights w_n are usually uniform, and BLEU-4 (1-gram to 4-gram) can be seen in equation 4. BP, which penalizes too short translations, is shown in equation 5, where c represents the length of the candidate translation and r is the effective reference length (the reference length closest to c).

$$W_n = \frac{1}{4}, n = 1,2,3,4 \quad (4)$$

$$BP = \begin{cases} 1 & if\ c > r \\ e^{(1-r/c)} & if\ c \leq r \end{cases} \quad (5)$$

III. Result and Discussion

A. Answer Generation Performance

This study implements a RAG system using three main language models: Gemma, LLaMA, and Qwen. These three models are integrated into a RAG pipeline consisting of three main stages: indexing, retrieval, and additional generation. The dataset consists of court decisions processed into vectors using embedding techniques. Performance evaluation was conducted by comparing the model's output with the ground truth using two primary metrics: ROUGE (ROUGE-1, ROUGE-2, and ROUGE-L) and BLEU.

According to Table 1 and Figure 5, Qwen achieves the highest score across all ROUGE variants. Figure 5(a) shows

that the ROUGE-L score for the Qwen model has the highest and most varied distribution, with a median of around 0.23 and a fairly wide range approaching 1.0. This indicates that Qwen can produce output with a higher degree of content overlap with the reference text than other models. The LLaMA model has a lower median score (~0.1). Meanwhile, the Gemma model performs the lowest, with a distribution skewing toward zero. This illustrates the Gemma model's difficulty in maintaining consistent word order with the reference.

TABLE I
MODEL PERFORMANCE COMPARISON

Model	Rouge-1	Rouge-2	Rouge-L	Bleu
Llama 3.2	0.1940	0.1034	0.1574	0.0592
Gemma 3	0.0730	0.0123	0.0521	0.0062
Qwen 3	0.3827	0.2773	0.3317	0.2162

In Figure 5(b), BLEU shows the performance of models with similar patterns. Qwen has the highest score, with a median of around 0.12 and a wider range of scores, up to 0.9. This indicates that Qwen is not only good at retaining key words and phrases but also excels at accurately preserving word order. The LLaMA model shows average performance, with a median of ~0.05, while Gemma shows very low scores with a narrow distribution. A notable aspect of Gemma's performance is its limited ability to generate summaries with sentence structures consistent with the source text.

Meanwhile, Figure 5(b) indicates that BLEU shows a similar pattern. The Qwen model again ranked the highest, with a median of around 0.12 and a wider range of scores, up to 0.9. This indicates that Qwen is not only good at retaining important words/phrases but also superior in maintaining accurate word order. The LLaMA model exhibited average performance, with a median of ~0.05. At the same time, Gemma showed very low scores with a narrow distribution, highlighting its limitations in producing translations or summaries with sentence structures consistent with the reference.

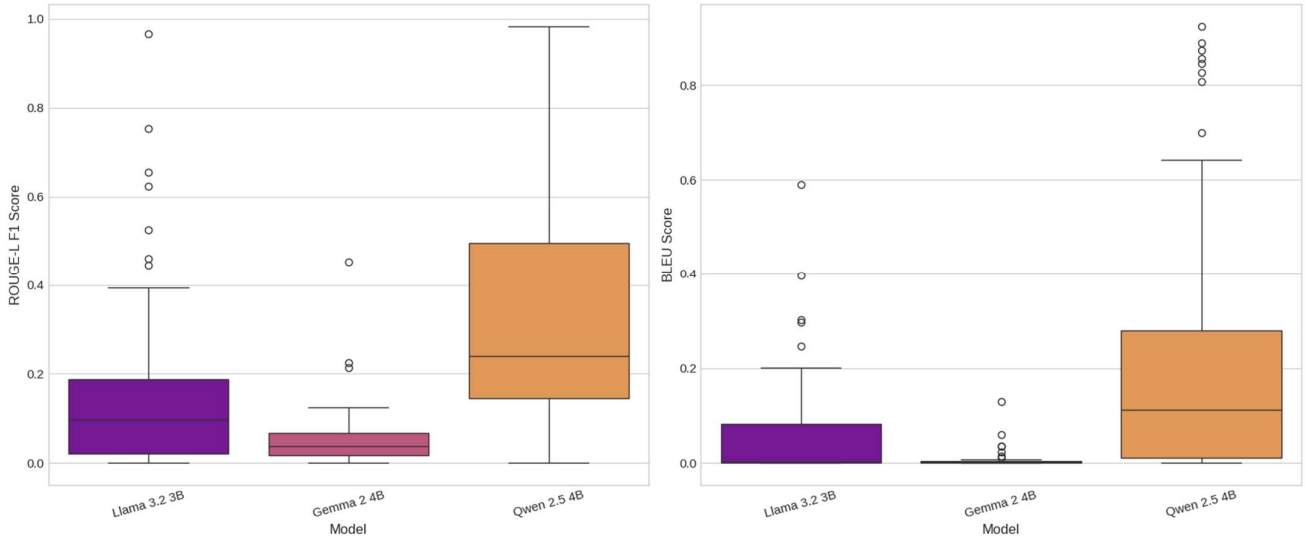


Fig. 5 Score distribution (a) ROUGE-L (Left) and (b) BLEU (Right)

Overall, these two visualizations demonstrate that integrating RAG with Qwen yielded more relevant and fluent output than LLaMA. In contrast, Gemma tended to lag in output quality, particularly in content (ROUGE-L) and word order (BLEU).

B. Retrieval Performance of Supporting Evidence

Evaluation of retrieval performance, as shown in Figure 6, indicates that BM25-based retrieval consistently outperforms dense retrieval across all reported ranking metrics. These results indicate that, for retrieval tasks involving Indonesian court decisions, lexical matching is more effective than semantic retrieval in identifying relevant legal passages. BM25's superiority in retrieval is attributed to the highly standardized structure of court documents, where recurring legal terms, fixed field names, and recurring procedural phrases make exact word matching highly informative. This confirms that the retrieval task in this dataset relies heavily on recognizing explicit legal expressions rather than inferring broader semantic similarities. [36], [37].

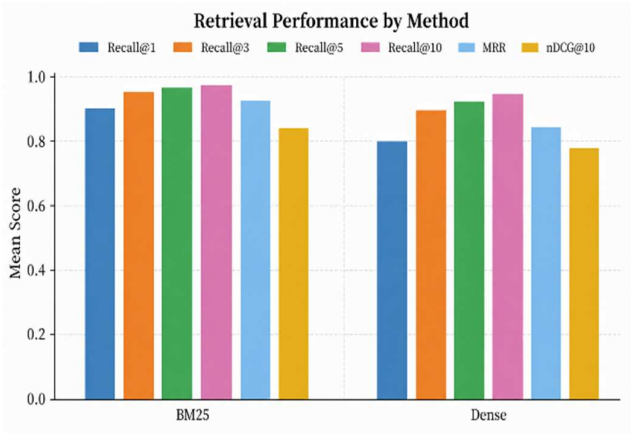


Fig. 6 Retrieval performance comparison between

C. Ablation and Baseline Analysis

1) Chunk Size and Top-k Ablation:

Figure 7 shows that retrieval performance improves as the chunk size and top-k increase. For the BM25 retrieval model, the highest average recall was achieved with a chunk size of 2,500 characters and top-k = 8, reaching 0.99. Dense retrieval follows the same trend but achieves a lower maximum recall of 0.95 under the same configuration. These results indicate that larger chunks preserve a more complete legal context along with key components of court rulings. This is because relevant information is often spread across several interconnected sentences rather than appearing in a single short phrase.

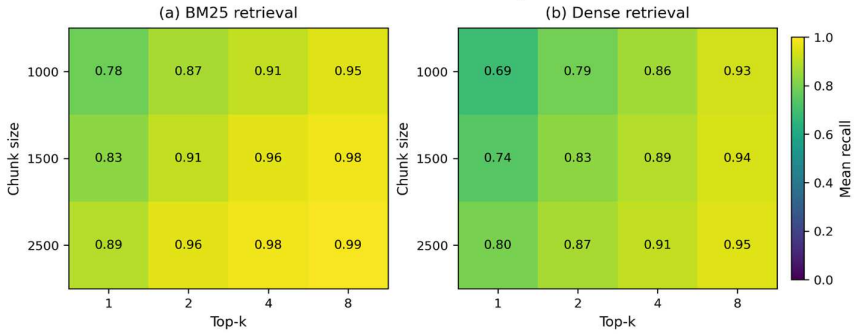


Fig. 7 Ablation analysis of retrieval recalls across chunk sizes and top-k values: (a) BM25 retrieval and (b) dense retrieval.

Increasing the number of selected chunks has also been shown to increase the likelihood that relevant portions of the decision will be included in the results. BM25 remains superior to dense retrieval in most configurations. These findings reinforce previous results indicating that lexical matching is highly effective for Indonesian court decisions. Court rulings often contain standard legal terms, recurring field names, and fixed procedural language. This ablation study supports the use of larger chunks in retrieval configurations, with BM25 being the most effective retrieval method for the current LQA task.

2) Baseline Comparison:

The aggregated results for answer quality across all models and baseline settings are presented in Table II. The results show that RAG-based configurations consistently outperform non-RAG LLM settings across F1 score, semantic similarity, ROUGE-L, and BLEU. Among all configurations, the combination of Qwen 3 with BM25 RAG achieved the highest overall answer quality. LLaMA 3.2 benefited from RAG, although its overall scores remained lower than those of Qwen 3. Gemma 3 performed the worst in most settings. Gemma 3 seemed to struggle to produce accurate legal answers given the provided context. The results of the answer quality matrix in Table 2 also demonstrate that the BM25 Retrieval-Only baseline achieves high semantic similarity because it directly returns the retrieved evidence. However, the ROUGE-L and BLEU scores for BM25 Retrieval-Only are low, indicating that retrieval alone is not sufficient to generate concise final answers. The findings in this matrix indicate that effective LQA requires accurate retrieval combined with a robust generative model [32], [36].

TABLE II
AGGREGATE ANSWER-QUALITY METRICS BY MODEL AND BASELINE

Model	Baseline	Token F1	Semantic Sim	ROUGE-L	Bleu	Unsupported Ratio	n
Gemma 3	BM25 RAG	0.0331	0.0442	0.0294	0.0120	0.9967	1728
Gemma 3	Dense RAG	0.0243	0.0348	0.0202	0.0074	0.9928	1728
Gemma 3	Non-RAG LLM	0.0008	0.0010	0.0006	0.0000	0.9994	1728
LLaMA 3.2	BM25 RAG	0.1914	0.2348	0.1884	0.0736	0.9973	1632
LLaMA 3.2	Dense RAG	0.1353	0.1842	0.1330	0.0386	0.9983	1631
LLaMA 3.2	Non-RAG LLM	0.0425	0.0607	0.0422	0.0060	0.9685	1620
Qwen 3	BM25 RAG	0.2724	0.3364	0.2625	0.1143	0.9896	1725
Qwen 3	Dense RAG	0.2150	0.2738	0.2042	0.0765	0.9889	1724
Qwen 3	Non-RAG LLM	0.0420	0.0559	0.0417	0.0075	0.9658	1725
BM25 Retrieval-Only	Retrieval only	0.0395	0.8610	0.0275	0.0086	1.0000	1728

D. Error Analysis

Distribution errors can be used to reveal different failure modes in various settings [38]. Figure 8 summarizes all of these patterns. Based on the results obtained in this study, rejection dominates the non-RAG output. Meanwhile, the Dense RAG and BM25 RAG outputs are fairly balanced. This means that when a model without RAG is uncertain or lacks sufficient information, it tends to refuse to answer the question. This study also identified a category of unsupported generation. This condition indicates that even though context is available, the model is still prone to generating statements that are not fully based on the retrieved evidence. In this category, errors in BM25 RAG and Dense RAG are quite dominant. Meanwhile, repetition errors are dominated by BM25 Retrieval-Only. This is because the extracted portion is returned without summarizing the answer. As for retrieval misses and wrong field boundaries, these errors occur less frequently across all models.

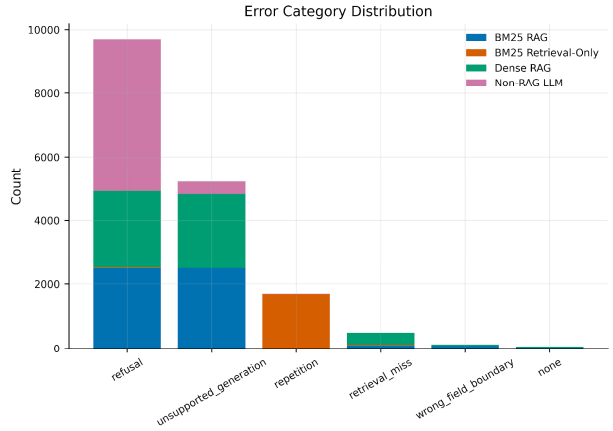


Fig. 8 Distribution of error categories across baselines.

E. Field-Level Performance Across Legal Categories

Figure 9 shows a comparison of the performance of three models (Qwen, LLaMA, and Gemma) on the five legal categories with the highest ROUGE-L values, including "*tuntutan*", "*petunjuk_bb*", "*amar_putusan*", "*penahanan*", and "*identitas_terdakwa*".

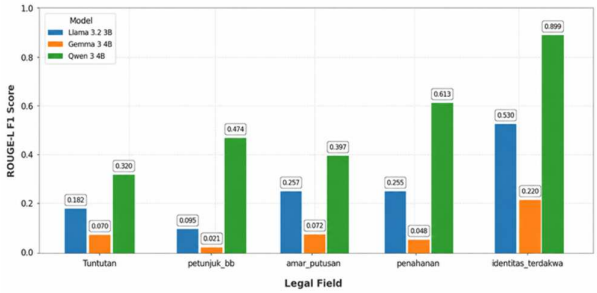


Fig. 9 Top five performing field.

In the legal verdict, the defendant's identity is presented in a standard format: name, age, address, occupation, and marital status. This consistency in language patterns makes RAG models easier to retrieve and produce, given the low linguistic variation, and the documents show many similarities. The identity of the defendant often appears throughout the document, from the beginning (case profile) to the judge's consideration. Due to its recurring appearance, information retrieval becomes more effective, improving query matching and providing significant advantages to the model.

The test results show that Qwen consistently performs better across all categories, with the strongest performance in the "*identitas_terdakwa*" category (0.899), far surpassing LLaMA (0.530) and Gemma (0.229). This result proves Qwen's ability to capture explicit structural information, including details of the defendant's identity, written in a standard format in the verdict. LLaMA and Gemma have not been able to handle structured and context-rich legal texts.

A comparison of the various metrics in this study shows that each evaluation method captures different aspects of performance. ROUGE emphasizes the overlap between words and phrases, making it useful for assessing content relevance. BLEU provides tighter measures of word-sequence accuracy and output smoothness. Retrieval-specific metrics such as Recall@k, MRR, and nDCG@10 complement these lexical metrics by directly measuring whether the system retrieves relevant evidence before creation. Overall, these results suggest that LQA evaluations are necessary to assess answer quality and retrieval effectiveness, as high lexical overlap alone does not fully indicate whether a model has taken and used correct legal evidence.

These findings reinforce the research finding that the effectiveness of RAG is not determined solely by model size. The effectiveness of the RAG is also strongly influenced by the suitability of the information retrieval strategies, the model's capacity, and the linguistic characteristics of each legal category. Categories with a clear structure, standard format, and consistent narrative patterns, such as "*identitas_terdakwa*" tend to result in higher evaluation scores because the information needed is easier to find and reconstruct. In contrast, categories that involve more diverse language, longer narratives, or interpretive elements, such as "*deskripsi_masalah*" and "*pakar*" remain more challenging because they require deeper contextual understanding and stronger semantic representations.

Therefore, the success of RAG in the field of law depends on the alignment between the linguistic complexity of the legal text, the retrieval method, and the generative capabilities of the chosen model. The integration of the Qwen 3 model with the BM25 in the RAG offers the best overall performance; LLaMA 3.2 provides an alternative with balanced performance. Gemma 3 is more limited to this task. More broadly, the development of RAG-based LQA systems does not depend solely on larger models, as it seems that simply computing parameters for each item is not a substitute for thinking. It must also consider the retrieval design, pre-processing strategy, grouping and enumeration methods, top-k selection, attribution mechanisms, and fine-tuning approaches tailored to the structure and language of the various categories of legal documents.

IV. Conclusion

The overall results of this study show that RAG generally improves the relevance and accuracy of large language model-generated answers in the LQA domain. Evaluations show that there is an improvement when RAGs are assessed against stronger benchmarks and retrieval-specific metrics, rather than relying solely on lexical overlap metrics. Compared to the Non-RAG LLM generation, the combination with the RAG of both BM25 RAG and Dense RAG achieved a higher answer quality score, while BM25 Retrieval-Only demonstrated that relevant legal evidence can often be retrieved successfully, although raw retrieval alone is not enough to produce a complete final answer.

Among the 2 information retrieval methods tested, the BM25 performed best across all retrieval test criteria. This condition shows that the retrieval of lexical information is well-suited for court decisions in Indonesia, whose wording often follows standard legal patterns. Further ablation studies showed that larger chunks and larger top-k values increased recalls. The fault analysis shows that the main remaining limitation is not widespread information retrieval failures. Still, rejection behavior and unsupported generation, suggesting that future improvements should focus not only on the quality of information retrieval but also on strengthening the answer and control generation.

Regarding model performance, further research must address the high levels of refusal behavior and unsupported generation by increasing response control in generative models and applying stronger reasoning. Another experiment that could be developed is related to the exploration of hybrid retrieval that combines the advantages of lexical retrieval with semantic retrieval for generative quality improvement. From the legal domain, it is also necessary to expand to less structured types of cases. Optimization and adaptation to the latest, most efficient LLM models are also needed to achieve quality and efficiency.

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